

ZeroDrift — how our demo actually works

Internal team explainer. Not for submission — this is so every one of us can answer any judge question.

The one-line pitch: Laser links need two telescopes pointed at each other to micro-radian class accuracy. Our software finds the other terminal's beacon and steers a virtual camera to keep it centred — the full coarse-alignment stage, built and measured, no optical hardware.

Golden rule on stage: never claim a number you cannot regenerate live. Every figure in the pitch deck comes from a logged run in the repo.

What is the beacon?

Physically: every FSOC terminal carries a small wide-beam LED/laser 'beacon' so the OTHER side can find it before the narrow data laser is aligned. It is a lighthouse for the handshake.

In our simulator: a moving point light source with real physics — brightness falls with distance squared, photon (shot) noise, and it BLINKS at 4 Hz on purpose.

Why blink? Stars and sun glints do not blink at 4 Hz. Our Goertzel filter measures the 4 Hz power in each candidate; a star scores ~0 and is rejected by a hard gate. That is how we never lock the wrong target (0 wrong locks in 64 runs).

Why 4 Hz exactly? Camera runs 30 fps → Nyquist limit 15 Hz. Blink must be well below that or the modulation aliases away. 4 Hz gives ~7 samples per blink cycle.

Where does the beacon COME FROM? (it is NOT an animation)

The pipeline each frame: real ISS orbit data (SGP4/TLE) → beacon's true direction in the sky → projected through a pin-hole camera model → photons on a 640×480 sensor → noise added (read noise, dark current, hot pixels, turbulence blur) → THAT image is all the tracker ever sees.

Hidden ground truth: the simulator knows the beacon's true position; the tracker does NOT. Truth is used only afterwards, to score the tracker. Like an exam: the examiner has the answer key, the student does not.

Judges can verify: the GUI has a 'show ground truth' toggle (green marker). We keep it OFF while tracking, then flip it ON to reveal the tracker was right. This is the moment that proves it is not staged.

Same seed → same run: every random draw is seeded. Run it twice, get identical frames. That is why the campaign and the deck numbers are reproducible.

Is the display 'just for show'?

No — the display is a window, not the machine. The tracker writes one telemetry record per frame (state, error, detections, timing). The GUI, the report and the website all just READ that record. Nothing on screen is typed in or animated by hand.

Desktop app: runs the real engine live — you can inject turbulence, add decoys, change blink rate DURING a run and watch it fight and recover.

Website (zerodrift-fsoc-pat.netlify.app): replays a LOGGED run of the same engine, frame by frame, from a JSON recording in the repo (docs/media/telemetry_run.json). Its banner says exactly that. It is for judges' phones; the app is for the stage.

If a number on the site and the deck ever disagree, we lose credibility instantly — that is why both now read from the SAME recording.

The five boxes (say it in 40 seconds)

VIRTUAL WORLD makes the sky: stars, clutter, decoys, turbulence, a beacon on a real orbit.

VIRTUAL CAMERA turns it into 30 frames/s of noisy pixels through a pan-tilt mount with 40 ms command delay.

PERCEPTION cleans each frame (top-hat filter), finds bright spot candidates (CFAR), locates them to 1/10th pixel.

DECISION asks 'which one is the real beacon?' — blink gate + AI verifier — and tracks it with an IMM Kalman filter.

CONTROL predicts 40 ms ahead (Smith predictor) and sends 30 pointing commands/s back to the mount. Loop closed.

SEARCH → ACQUIRE → TRACK → COAST → REACQUIRE: the state machine. COAST = beacon blinked off, we keep pointing from prediction. That is why detection duty ~50% but lock ~98%.

Numbers we quote — and their source (memorise this table)

1.27 s acquisition · 98.3% lock · 198 μ rad median · 100% in-FOV: the recorded ISS-pass run — docs/media/telemetry_run.json (what the website replays and the slide-2 chart plots).

0/64 wrong-target locks · worst case 4.0 s acq, 91.6% lock: 64-run Monte-Carlo campaign — runs/mc-leo/summary.json.

$\approx$ 20 ms of the 33 ms frame budget: runs/iss_pass.report.json processing p50.

99.3% vs 14.7% link availability: link-budget model, with vs without coarse alignment — src/fsoc_pat/linkbudget.py.

73/73 tests · 5,000+ lines: pytest suite and wc -l, run them live if asked.

AI verifier AUC 0.957 vs 0.900 classical: docs/technical_report.md — and note we benchmarked AI against the classical method, we did not just sprinkle AI.

Judge Q&A — likely questions, one-line answers

What is a μrad ? One micro-radian tilts the beam ~ 1 mm sideways per km of distance. Our 198 μrad median at 550 km class geometry keeps the beacon well inside the $\pm 3^\circ$ field.

Why only coarse alignment? The PS asks for it: coarse gets the beacon into the field of a fast fine-steering mirror; the FSM (kHz) does the last microradians. Two-stage is how real terminals work.

Why not just track the brightest pixel? Sun glints and decoys are brighter. Identity comes from the blink signature, not brightness.

Where exactly is the AI? A 497-parameter neural verifier we trained from scratch (no dataset downloaded — we generate labelled data from our own simulator) that confirms a track is beacon-like. Gate stays classical; AI adds confidence.

What about the 40 ms delay? A Smith predictor: an internal model of the mount runs 40 ms ahead, so commands aim where the target WILL be.

Hardware? The problem statement explicitly wants algorithm development without optical hardware — that is the point of a virtual testbed. Everything runs on the judges' own laptop.

Is the web demo live? It is an honest replay of a logged run of the real engine — and we will run the engine live right here, change the physics mid-run, and let you watch it recover.

Stage roles & the 60-second open

Driver (laptop): starts leo_pass_nominal, real-time ON, truth OFF. On cue: turbulence slider up, then truth ON.

Narrator: the 40-second five-boxes story while the app acquires. Then: 'notice it never saw the answer key — now we reveal it.'

Q&A anchor 1 (physics/controls): beacon, blink, Smith predictor, μ rad questions.

Q&A anchor 2 (AI/software): verifier NN, tests, reproducibility, packaging.

Opening line: 'Every number you will see is measured, and we will regenerate one for you live.' Then do it.

Never say: 'basically', 'obviously', 'it should work'. Say: 'measured', 'reproducible', 'watch'.